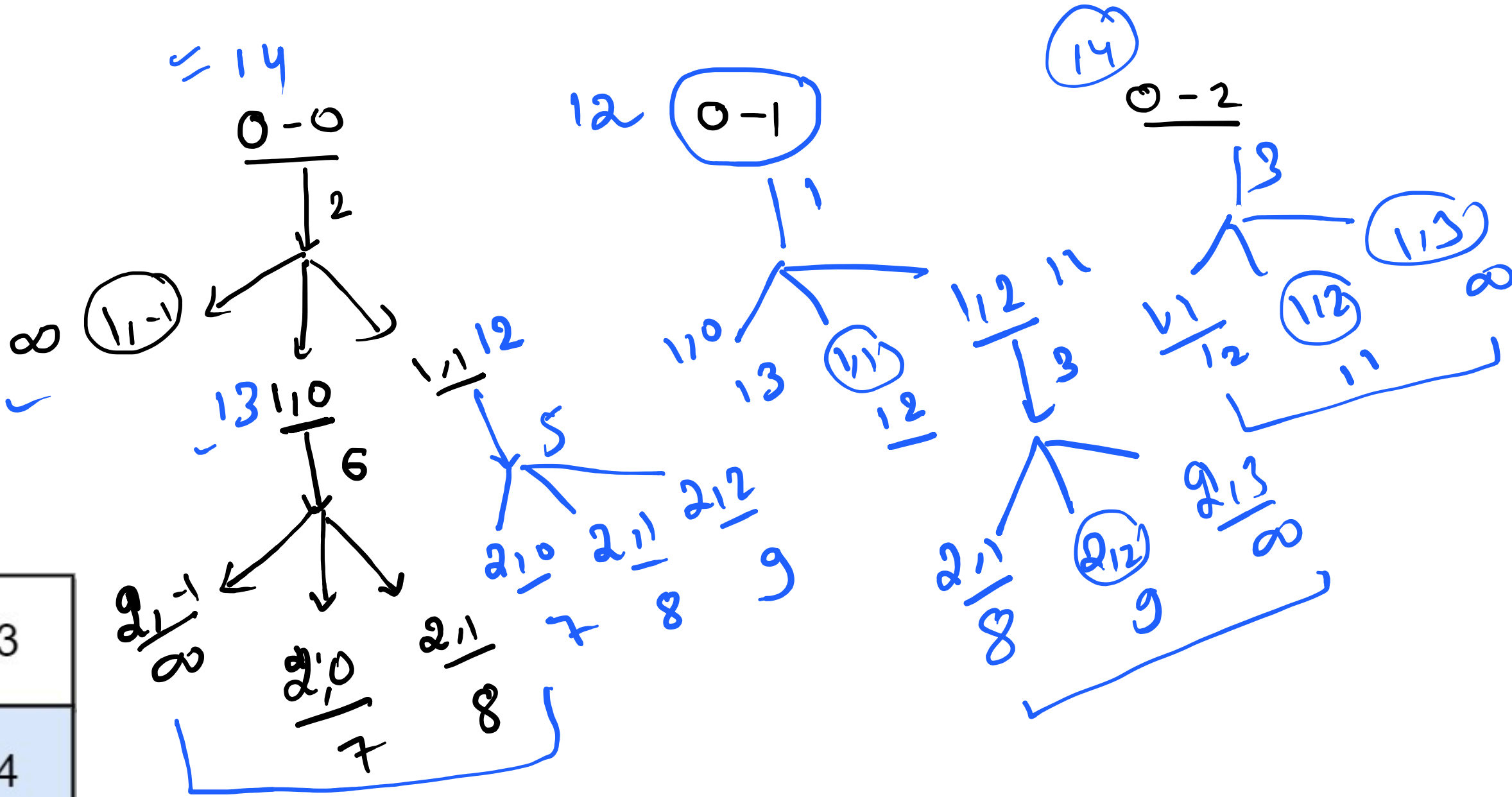
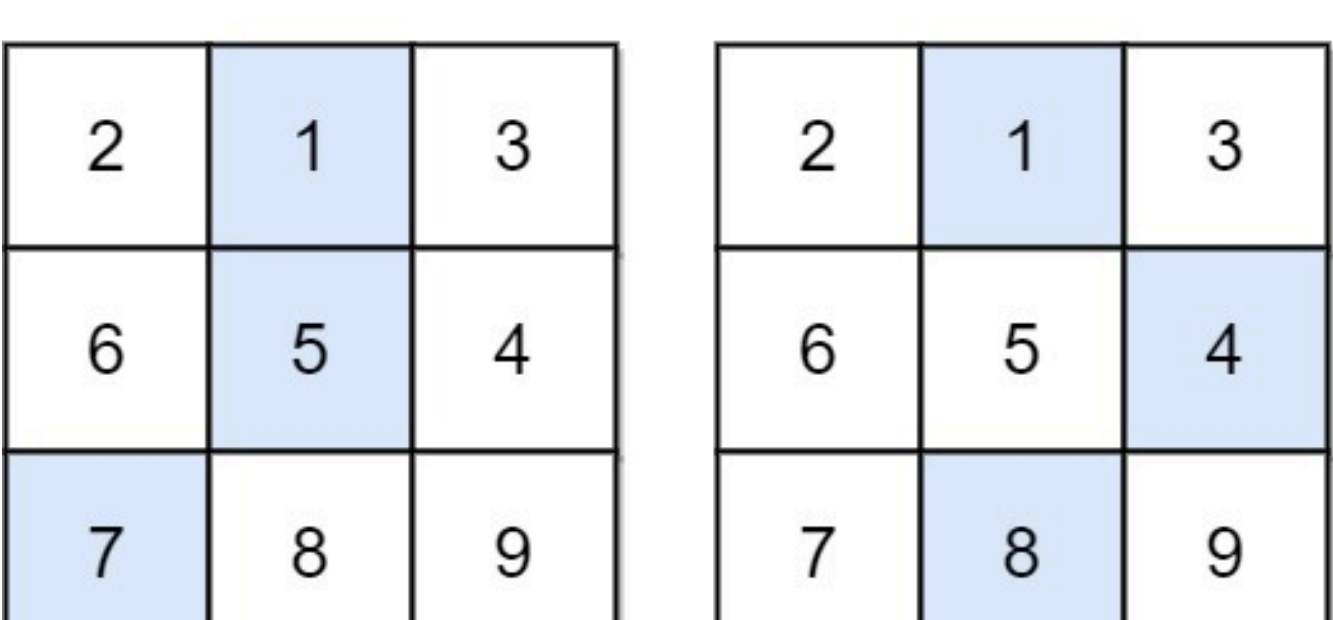
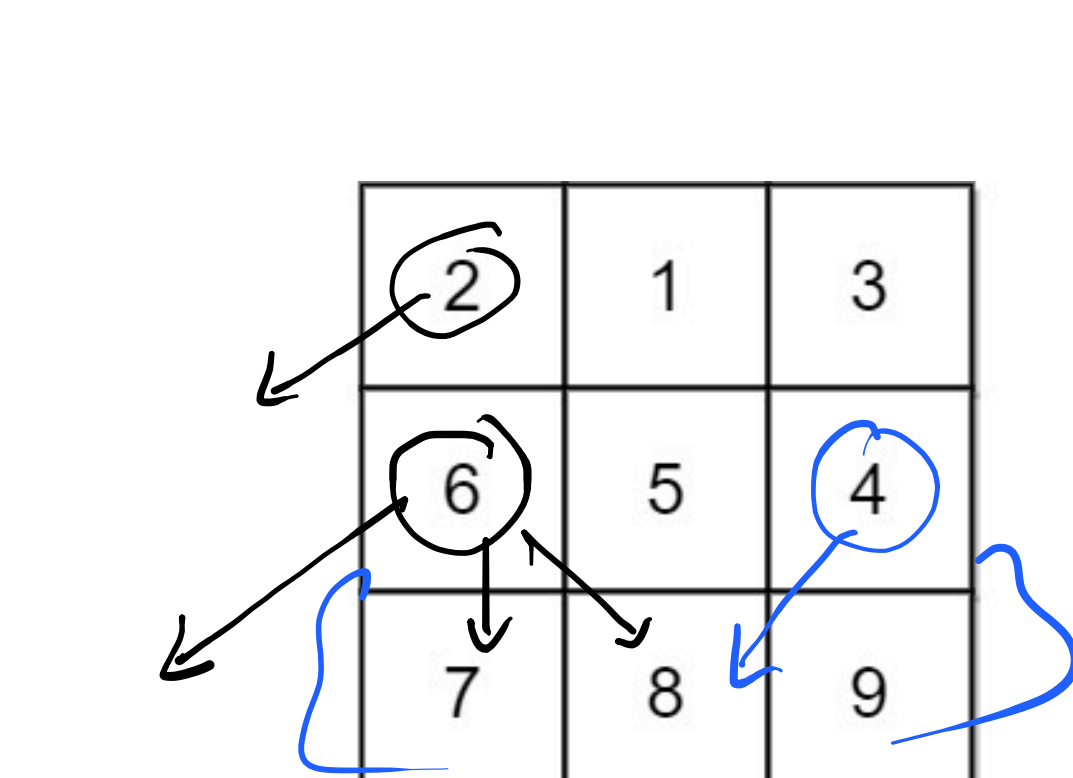
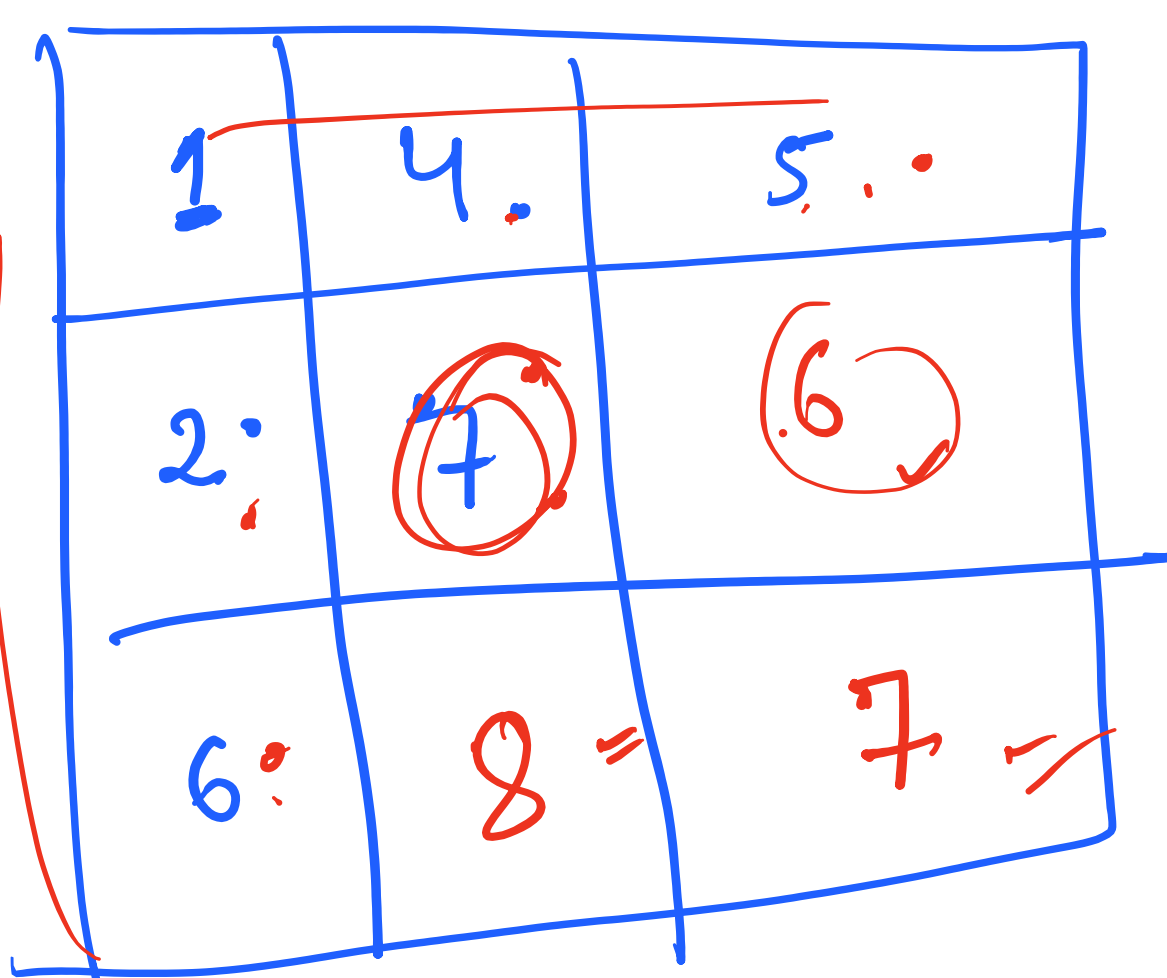
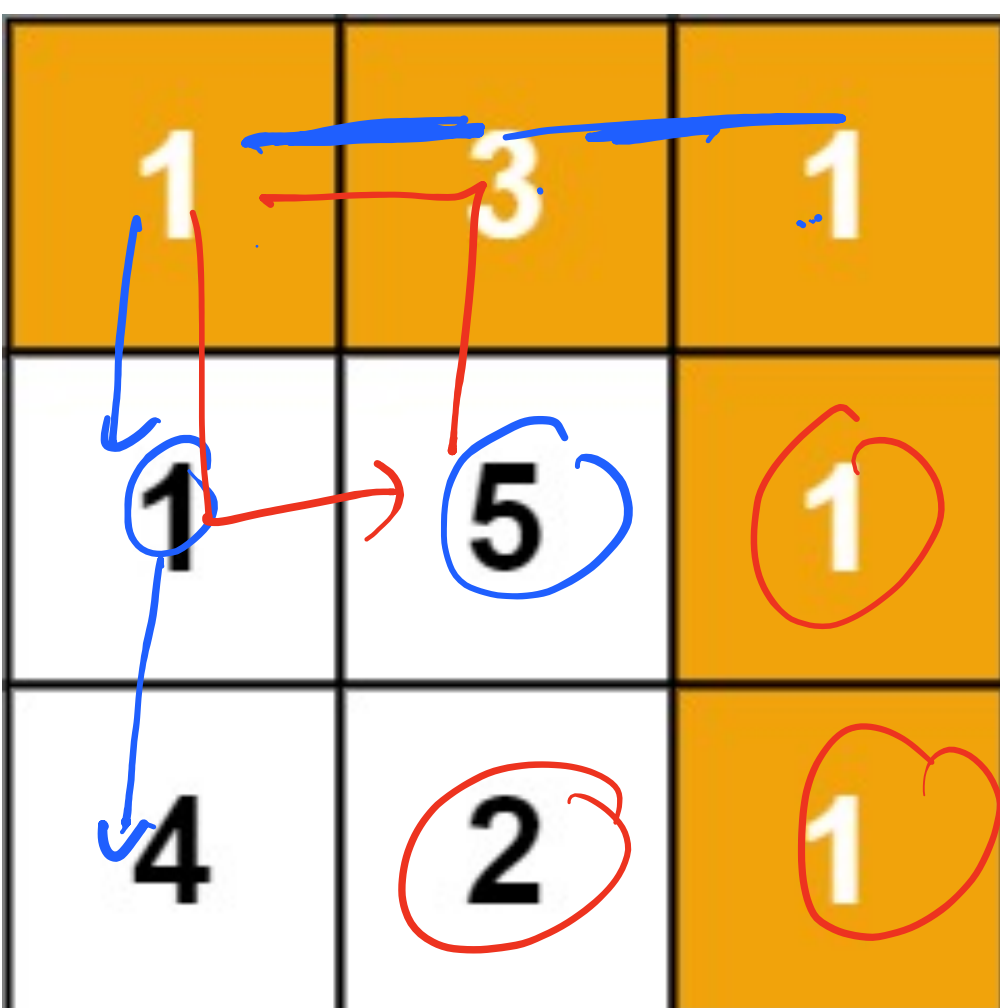
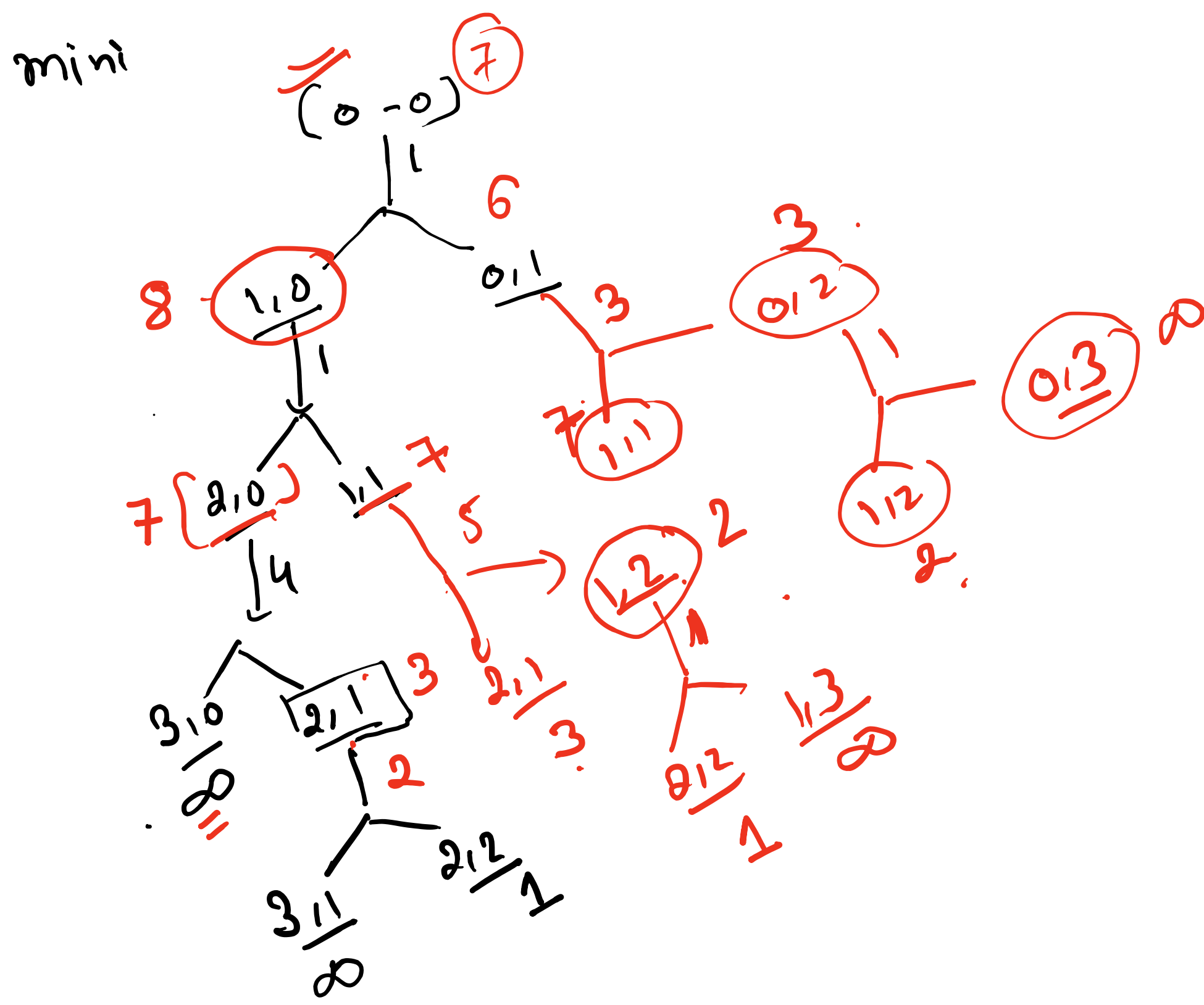
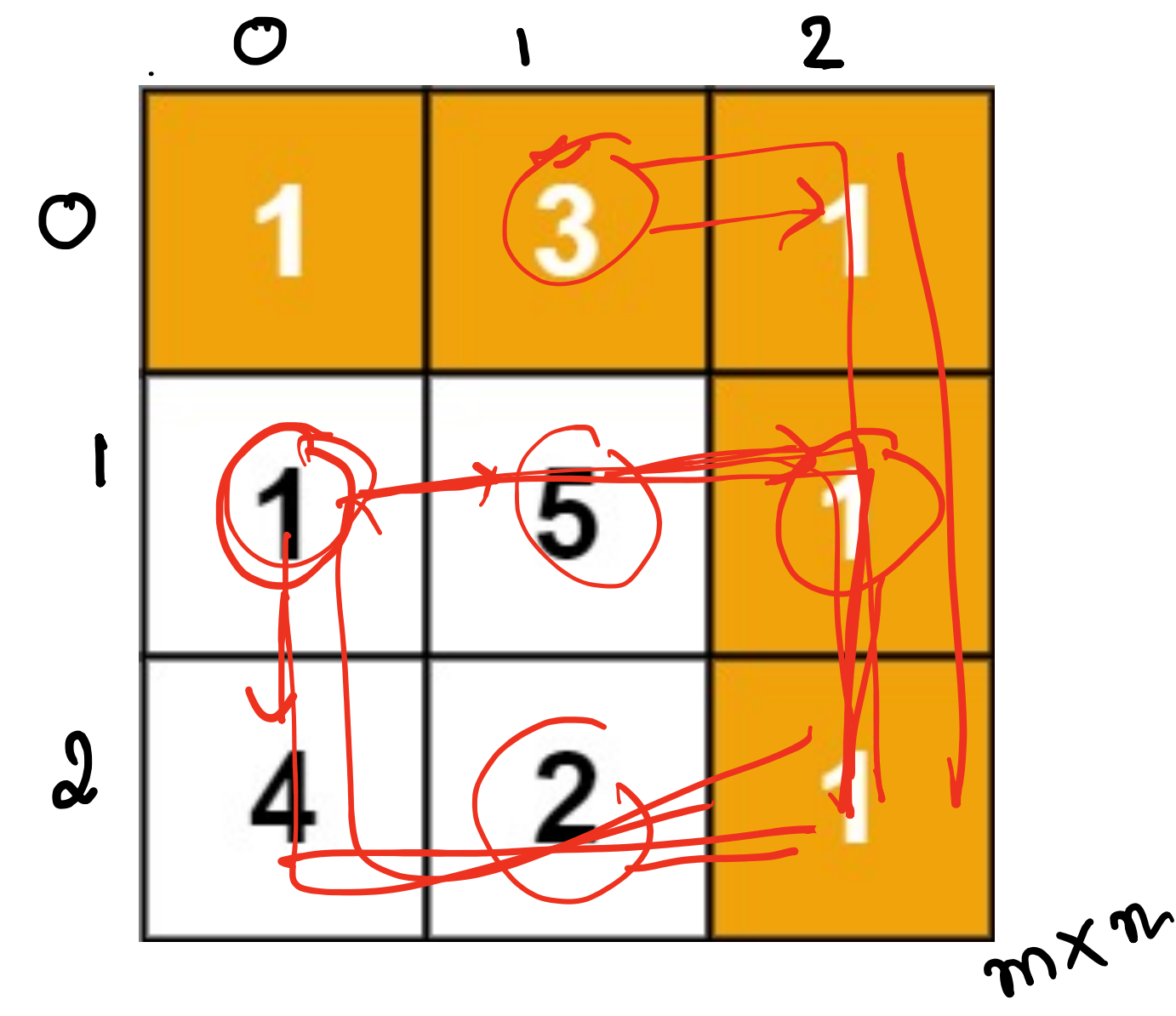
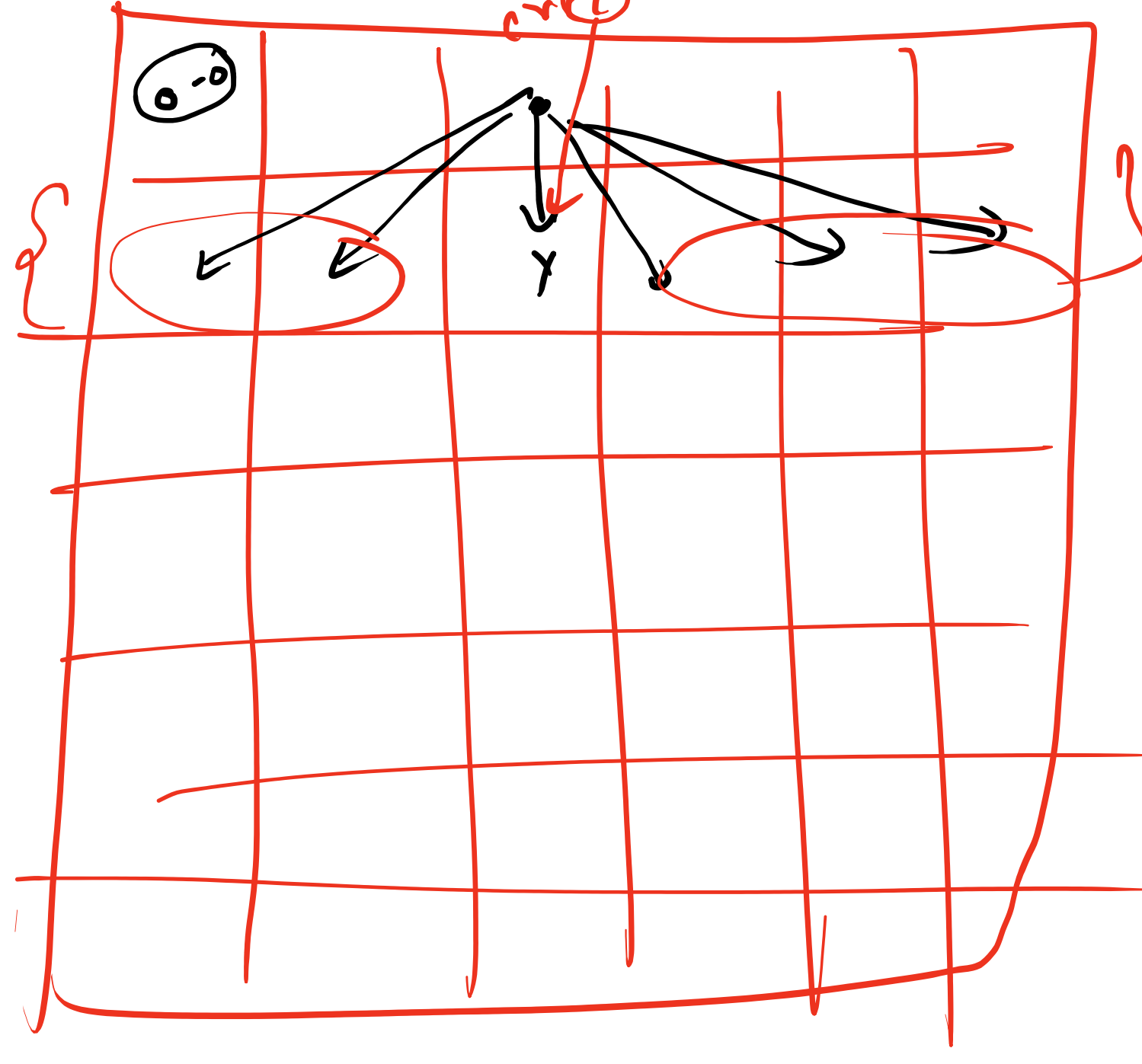
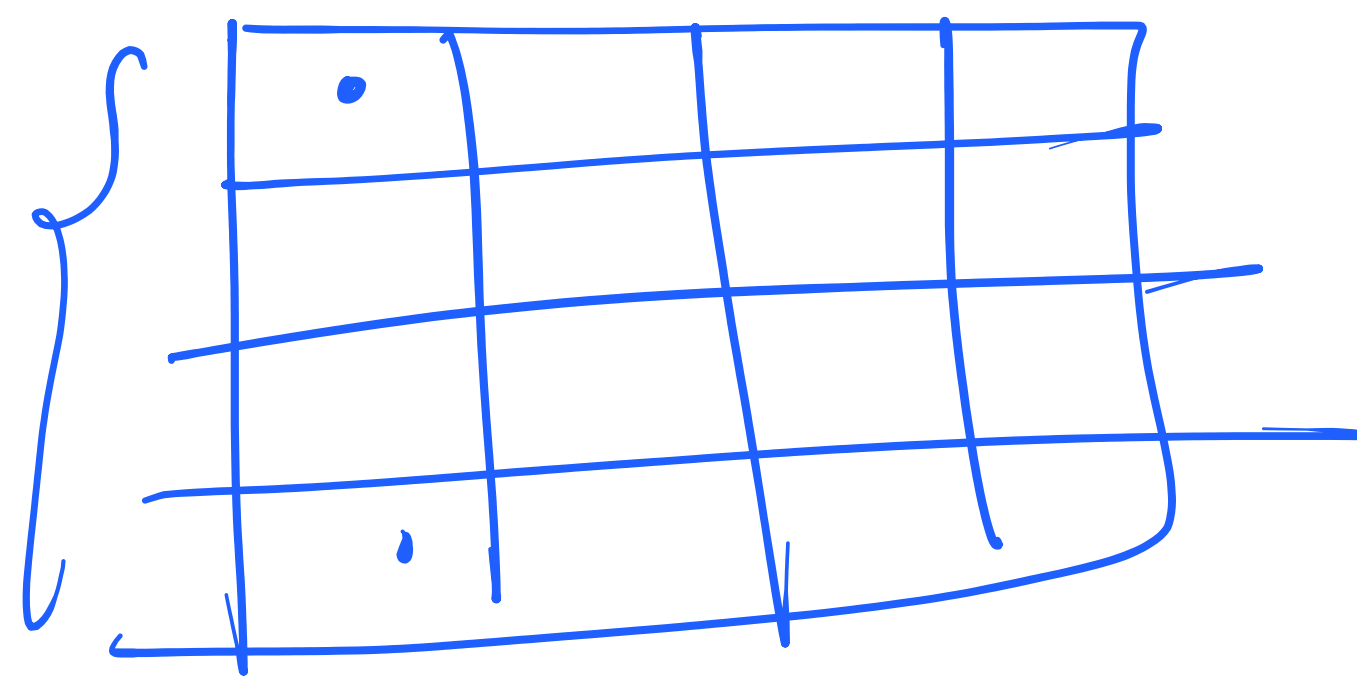


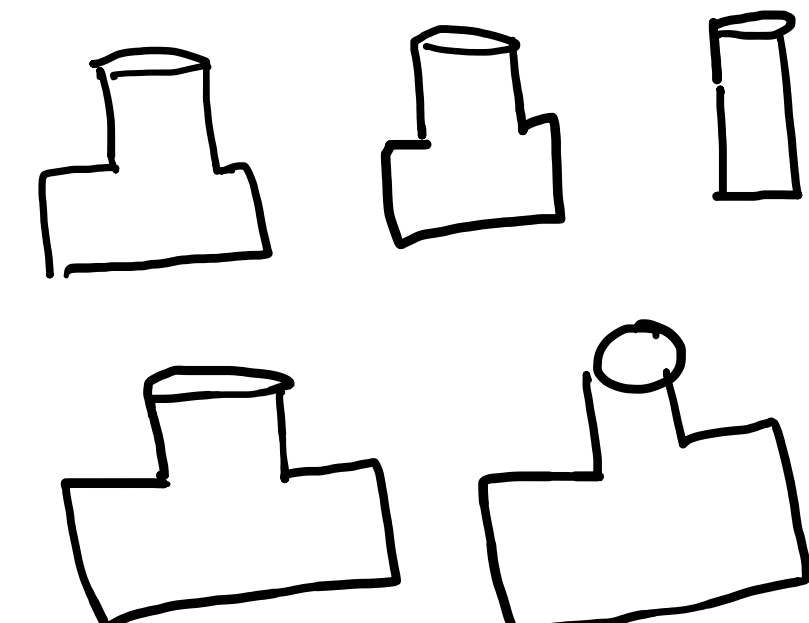


$$\begin{array}{r} 100 \times 100 \\ -1000 \\ \hline -100 \\ -100 \\ \hline -200 \end{array}$$

$$-400$$



Given n wines in a row, with integers denoting the cost of each wine respectively. Each year you can sell the first or the last wine in the row. Let the initial profits from the wines be $P_1, P_2, P_3 \dots P_n$. In the Y th year, the profit from the i th wine will be $Y * P[i]$. The goal is to calculate the maximum profit that can be earned by selling all the wines. Suppose, wine array denotes the initial cost of each wine in the first year. wine[] = [2, 3, 5, 1, 4]



$$\begin{array}{r} 2 \ 3 \ 5 \ 1 \ 4 \\ \downarrow \quad \downarrow \\ 2 \ 3 \ 5 \ 1 \ 4 \\ \underline{3 \ 5 \ 1 \ 4} \\ 2 \times 1 = 2 \quad 5 \times 4 = 20 \\ 3 \times 2 = 6 \quad 1 \times 4 = 4 \\ 4 \times 3 = 12 \quad 5 \times 1 = 5 \\ 1 \times 4 = 4 \quad 5 \times 5 = 25 \\ \hline 49 \end{array}$$

$$5 - \frac{(j-i)}{4} = 1$$

$$5 - \frac{(1-1)}{4} = 2$$

$$5 - \frac{(2-1)}{4} = 2$$

$$[2, 3, 5, 1, 4]$$

$$2 \times 1 + [3, 5, 1, 4]$$

$$4 \times 1 + [2, 3, 5, 1]$$

$$j - i + 1$$

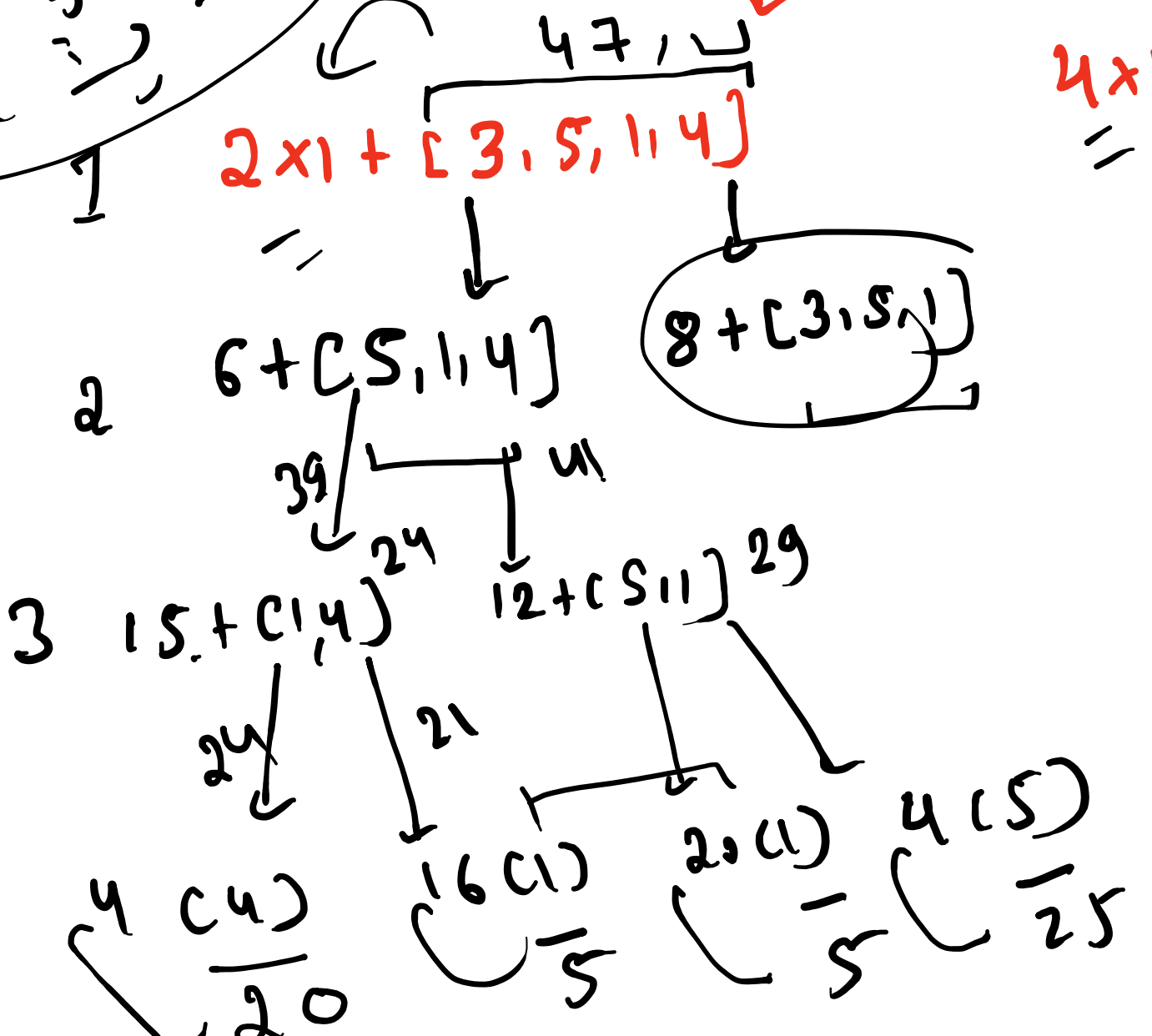
$$j - i + 1$$

$$n = 1$$

$$n - 1 = 2$$

$$n - 2 = 3$$

$$\max \left\{ \begin{array}{l} \text{wine}[i] \times Y + \text{RC}[i+1:j] \\ \text{wine}[j] \times Y + \text{RC}[i:j-1] \end{array} \right\}$$

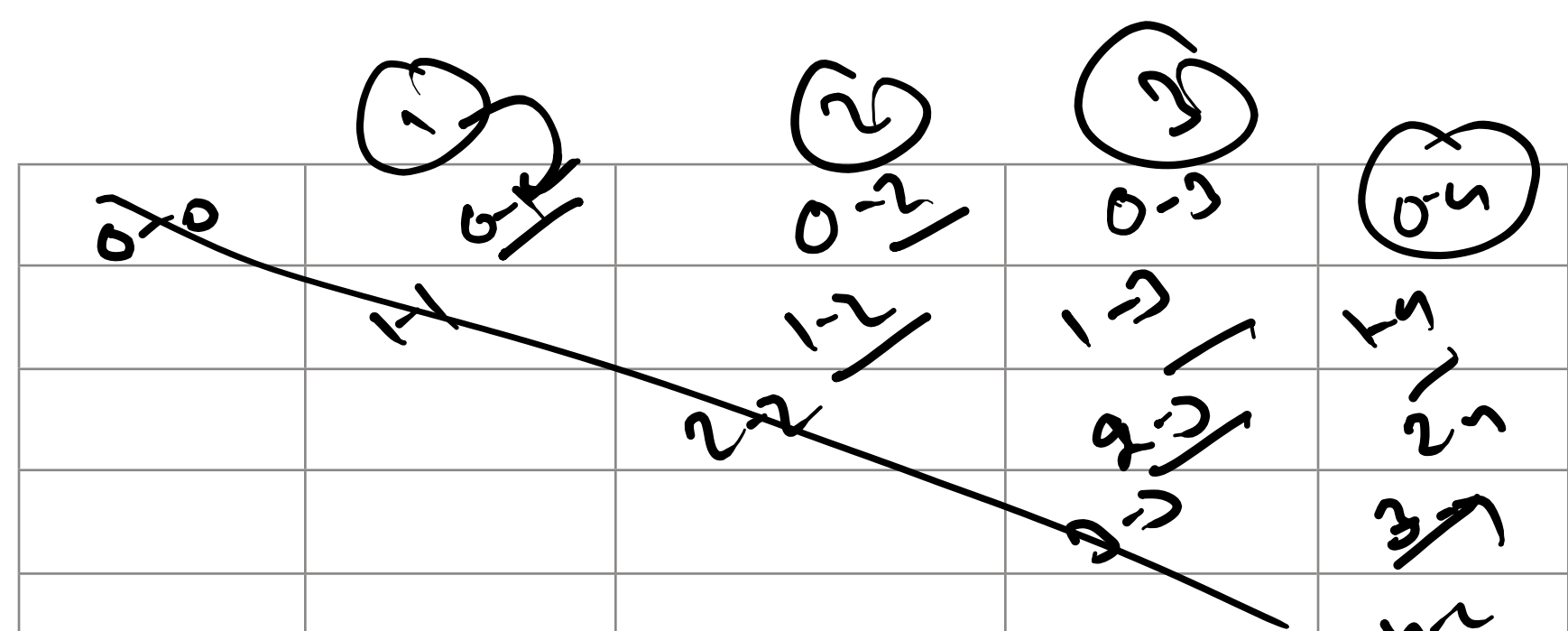
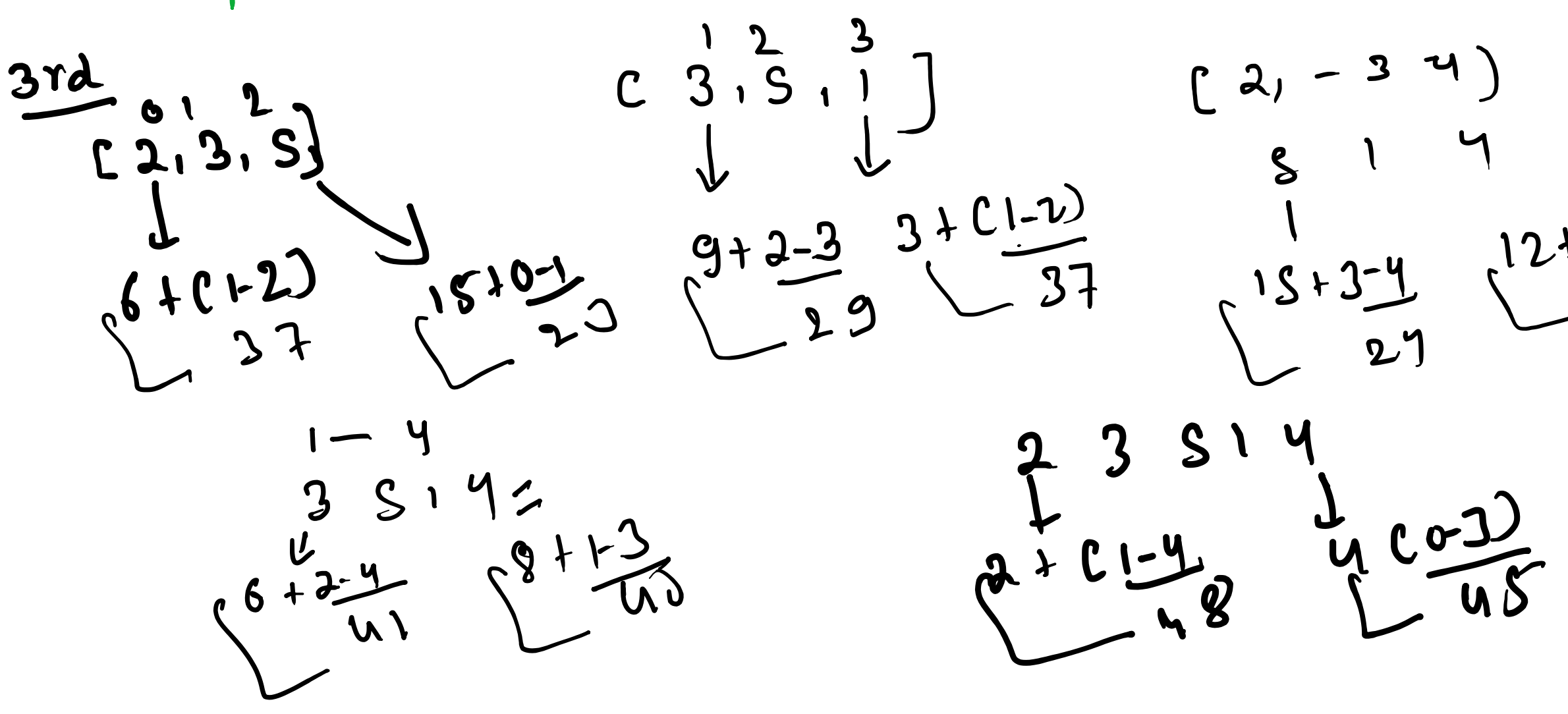
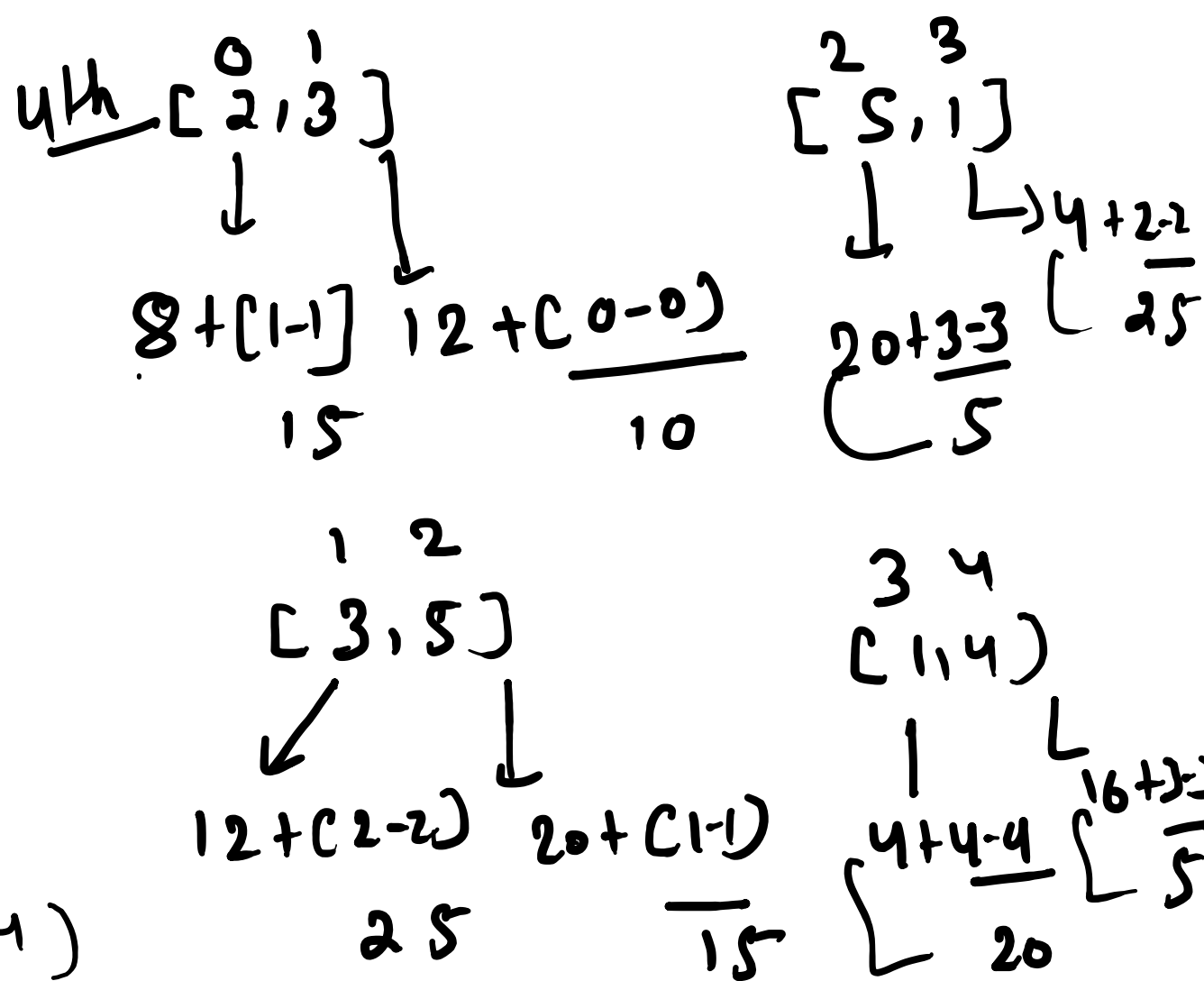


$$i > j$$

1

dp[i][j] (cost)

	0	1	2	3	4
0	2	10	23	43	45 ← 50
1	3		18	37	40
2	5			25	29
3	1				5
4	4				



Full get = 1 & n get

$$j - i = \text{gap}$$